An Obstruction Calculus for Viability under Incomplete Observation

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Abstract

Under perfect measurement the viability kernel — the set of states from which some feedback keeps the system within its constraints — is characterized by tangency conditions. Under incomplete observation the sufficiency direction has a canonical answer in Veliov's output-feedback regulation condition and the estimation-tube reduction. The complementary direction — certifying that no observation-based policy is viable — has lacked a comparable instrument.

This paper develops that instrument: a calculus of obstruction certificates, sound sufficient conditions for nonviability (equivalently, necessary conditions for observation-based viability), finitely checkable in the polyhedral, finite-fibre, and finite-horizon cases and closed-form elsewhere. They do not exhaust the complement of the epistemic kernel, the observation-based counterpart of the viability kernel; in finite systems backward recursion is complete. Five mechanisms are established, with a sixth exhibited under a policy-class restriction. The central common-action obstruction shows that when the safe controls of compatible states intersect emptily, no observation-based policy is viable, though every compatible state is individually viable under full information. The others are a finite-time exit certificate under an Isaacs-type drift condition, an epistemic-emptiness construction in which a constant observation merges states with incompatible admissible controls, a delayed-information obstruction with an explicit timing bound, and a certification limit: an exact observation-only certifier exists exactly when safe-set membership is constant on the observation fibres. The sixth, a certainty-equivalence trap, empties the epistemic kernel under a biased observation even when the perfect-information kernel is nonempty.

The calculus is positioned against barrier certificates and estimation tubes, and its consequences for monitoring and institutional design — observation timing, coarseness, aggregation, and bias — are drawn.

Keywords: viability theory; viability kernel; obstruction certificates; robust viability; epistemic viability; output feedback; sustainability governance; information structures

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1. Introduction

1.1 The problem

Viability theory characterizes the states of a constrained control system from which there exists at least one control keeping every future state within a constraint set (Aubin, 1991). For sustainability problems the constraint set is a set of floors — stock levels, service thresholds, safety margins — and viability is the formal counterpart of the requirement that a development path be maintained rather than merely optimized (Béné, Doyen, and Gabay, 2001; De Lara and Doyen, 2008; Doyen et al., 2012; Doyen and Gajardo, 2020). The theory under *perfect measurement* is complete in the relevant sense: the viability kernel exists as the largest viable subset, and Nagumo-type tangency conditions certify it.

Sustainability governance, however, operates under *incomplete observation*. Stocks are assessed at discrete review intervals. Indicators are coarser than the state. Some stocks are unobserved altogether. And the information that does arrive may arrive late. This paper addresses that side: under an incomplete observation structure, it develops instruments for certifying that no observation-based policy is viable — establishing infeasibility for reasons of information rather than of dynamics.

Floors in sustainability assessment are typically constraints on the productive base — the state of the system that yields the measured output — rather than on the output itself. Because a base can be drawn down while measured output is maintained, the signal that would reveal erosion is often the one that is coarse, postponed, or absent. Incompleteness of observation is therefore a structural feature of the certification problem, not merely a practical limitation; the sufficiency literature and the obstruction calculus of this paper address the same underlying indeterminacy rather than competing ones.

The question has a natural split. The *sufficiency* direction has a canonical literature. Veliov (1993) gives conditions under which an output-feedback regulation map exists when only incomplete and inexact measurement is available, and shows that under perfect measurement his condition reduces to the classical viability condition of Haddad (1981). The estimation-tube programme of Quincampoix, Cardaliaguet, and Saint-Pierre (see Cardaliaguet, Quincampoix, and Saint-Pierre, 2007) passes from imperfect information in the measurement space to perfect information in an estimation space, with equality of the corresponding value functions and a Dini-derivative characterization. In the control-verification literature, barrier certificates certify safety for continuous and hybrid systems (Prajna and Jadbabaie, 2004; Prajna, Jadbabaie, and Pappas, 2007), and converse results show that, under convex-duality conditions on density functions, the existence of a barrier certificate is also *necessary* for safety (Prajna and Rantzer, 2005), with necessary-and-sufficient characterizations for hybrid inclusions under mild regularity (Maghenem and Sanfelice, 2019).

What has been missing is a calculus of obstruction certificates. An obstruction certificate — a checkable witness that no observation-based policy exists, finite in the polyhedral and finite-fibre cases — is an argument that a prescribed class of observation-based policies fails, not because a particular policy is bad, but because the information structure leaves no room for any policy. In its polyhedral common-action and certification forms (Theorem 2 and Proposition 4) the argument is a finite, checkable test. The drift certificates of Theorem 1 and Theorem 3 are not finite objects — one exhibits an enforcement selection, the other bounds the timing uniformly over every policy. The two directions are complementary. Veliov's condition tells us when output feedback *can* work; the obstruction calculus tells us when it *cannot*. For the borderline cases it identifies which quantitative feature of the observation design — timing, coarseness, bias, aggregation — is responsible.

1.2 Contributions

Five obstruction mechanisms are developed, each with a complete proof, and one further mechanism is exhibited by construction:

1. **The finite-time exit certificate** (Theorem 1). Under a Dini-drift condition of Isaacs type — the disturbance chooses after the control — the disturbance can force exit from the safe set within an explicit time bound, for *every* admissible control of any information structure. This is the base obstruction: when it applies, the viability question is closed without any observation-theoretic argument.

2. **Epistemic emptiness by admissibility** (Proposition 1, a minimal construction). A constant observation merges states whose *admissible* controls differ: the state-dependent control set forces a single common action, and that action is unsafe at the boundary, so the unresolved initial observation fibre lies outside the epistemic kernel, even though every physical state is viable under full information. The mechanism is admissibility — the common-action obstruction of Theorem 2 at minimal size — and Example 1 gives the hidden-mode instance.

3. **The instantaneous common-action obstruction** (Theorem 2, Example 1). If the safe-control sets of the states compatible with the current information set intersect emptily, then no observation-based policy is viable, even though every compatible state is individually viable under full information. The failure is purely informational: no stochasticity, no estimation error.

4. **The delayed-information obstruction** (Theorem 3). If the information needed to distinguish safe from unsafe compatible states arrives only at time T_{obs}, and the disturbance can force constraint violation earlier than T_{obs}, then no observation-based policy is viable. The timing bound $\inf q/\varepsilon < T_{\mathrm{obs}}$ is stated explicitly: information may be accurate but arrive too late.

5. **The fibre certification criterion** (Proposition 4, Corollary 1). An exact observation-only safety certifier — a function of the observation that returns "safe" exactly on the safe set — exists if and only if safe-set membership is constant on every observation fibre. The certainly-safe set is the sound relaxation: the largest set of observations that can be labeled safe without further information.

6. **The certainty-equivalence trap** (Remark 3). Even an *injective* observation empties the epistemic kernel of a system whose perfect-information kernel is nonempty if the policy class is restricted to certainty-equivalence controllers that apply a fixed state-feedback law to the uncorrected observation. The kernel empties by restriction of the policy class, not by loss of information.

The calculus is organized by a single selector principle (Section 2.4): observation-based viability is the existence of one response — admissible, safe, and recursively viable — for every compatible state, and each certificate proves that a specific response correspondence is empty; the obstructions are nested in a ladder (Proposition 3). Four refinements complete the picture: a comparison-function form sharpens the exit-time bound (Remark 1); a uniform-margin condition connects the instantaneous and tube obstructions (Proposition 2); the timing obstruction has a sharp threshold form (Remark 2); and backward belief recursion is sound and complete on any finite horizon in finite systems (Theorem 4). The epistemic kernel is monotone in action sets, disturbances, information, and policy class (Proposition 5).

Throughout, "obstruction" means a *certified* failure. Theorem 1 and Theorem 3 each exhibit the violating constraint, an admissible disturbance, and the quantitative bound that make failure inevitable, for every policy in the prescribed class; Proposition 1 and Theorem 2 exhibit the incompatible admissible controls — the merged observation whose incompatible safe controls certify emptiness — with no quantitative bound claimed there; and Proposition 4 is the converse-side characterization of the certification limit itself. The calculus is positioned against the barrier-certificate literature (Section 6.2) and the estimation-tube programme (Section 6.3), and its consequences for the design of monitoring and indicators in sustainability governance are drawn in Section 6.4.

1.3 Related work

The viability theory background is Aubin (1991) and Aubin, Bayen, and Saint-Pierre (2011); approximation of kernels is due to Saint-Pierre (1994). Robust (disturbance-averse) viability is developed in Aubin and Frankowska (1990) and Frankowska (1989). Under incomplete measurement, Veliov (1993) gives the sufficiency theorem already mentioned; Quincampoix and Veliov (1994) treat viability with a priori unknown but observable parameters; and the estimation-set reduction is presented in Cardaliaguet, Quincampoix, and Saint-Pierre (2007). In verification, barrier certificates originate with Prajna and Jadbabaie (2004); their worst-case framework and converse are Prajna, Jadbabaie, and Pappas (2007) and Prajna and Rantzer (2005); hybrid necessary-and-sufficient characterizations are Maghenem and Sanfelice (2019). The sustainability application domain is represented by Béné, Doyen, and Gabay (2001), De Lara and Doyen (2008), Doyen et al. (2012), and Doyen and Gajardo (2020). To our knowledge, the obstruction certificates of Theorem 2, Theorem 3, and Proposition 4 — in particular the certification criterion of Proposition 4 and the timing bound of Theorem 3 — have not been stated in this form; the elementary facts on which they rest (quantifier commutation; Dini comparison) are classical.

1.4 Organization

Section 2 fixes the framework: the control system, the viability kernel hierarchy, observation structures, information states, and the safe-control correspondence. Section 3 develops the obstruction calculus (Theorem 1–4, Propositions 1–3, Remarks 1–2, and Example 1): the finite-time exit certificate, the admissibility and common-action obstructions, the delayed-information obstruction with its threshold form, the obstruction ladder, and finite-horizon completeness. Section 4 develops the certification limits (Proposition 4, Corollary 1, Remark 3). Section 5 reviews the sufficiency landscape, which this paper does not re-derive, with citations. Section 6 discusses the position of the calculus relative to barrier certificates and estimation tubes, states the monotonicity of the epistemic kernel (Proposition 5), and draws the governance consequences. Section 7 concludes. Proofs are complete in the main text; results whose proofs repeat established literature are cited in Section 5 rather than reproduced.

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2. Framework

2.1 The control system and the kernel hierarchy

This section fixes the mathematical objects used throughout the paper: the control system, the disturbance structure, the constraint set, and the four kernels (perfect-information, robust, epistemic, and institutionally restricted) that organize the analysis.

Fix a state space $X \subseteq \mathbb{R}^n$, a control set U, a disturbance set D, and dynamics

$$\dot{x}(t) = f(x(t), u(t), d(t)), \qquad u(t) \in U(x(t)), \quad d(t) \in D(x(t)),$$

with f continuous, U and D set-valued with closed graph and nonempty compact values, and admissible controls and disturbances taken to be measurable selections. Let $\mathcal{V} \subseteq X$ be a closed constraint set. A *causal policy* π is a map from the observation record to controls, respecting the specified information structure. The standard objects are:

- $\mathrm{Viab}(\mathcal{V}; U, \pi_{\mathrm{perf}})$: the **viability kernel** under full-information (state-feedback) policies — the largest closed subset of $\mathcal{V}$ from which there exists a state-feedback control keeping the trajectory in $\mathcal{V}$ for all time (Aubin, 1991).

- $\mathrm{RViab}(\mathcal{V})$: the **robust kernel**, the largest subset of $\mathcal{V}$ from which there exists a state-feedback policy keeping every trajectory — for *every* admissible disturbance realization — in $\mathcal{V}$ (Aubin and Frankowska, 1990).
- $\mathrm{ERViab}_{\mathcal{I}}(\mathcal{V})$: the **robust epistemic kernel** — the object of this paper (Definition 1). Its elements are *states of information*, not physical states (Section 2.3). The non-robust counterpart $\mathrm{EViab}_{\mathcal{I}}(\mathcal{V})$ is the non-robust contrast class (Section 2.3).

Projected to physical state space — with $\mathrm{proj}_{\exists}(\mathfrak{B}) = \bigcup_{B\in\mathfrak{B}} B$ the *existential* projection of a collection $\mathfrak{B}$ of information states, applied to the first two terms, whose elements are information states — the informational hierarchy reads

$$\mathrm{proj}_{\exists}\big(\mathrm{IRViab}_{\mathcal{J}}(\mathcal{V})\big) \;\subseteq\; \mathrm{proj}_{\exists}\big(\mathrm{ERViab}_{\mathcal{I}}(\mathcal{V})\big) \;\subseteq\; \mathrm{RViab}(\mathcal{V}) \;\subseteq\; \mathrm{Viab}(\mathcal{V}),$$

where $\mathrm{IRViab}_{\mathcal{J}}(\mathcal{V})$ is the institutionally restricted kernel of Section 6.4. The physical implication of epistemic viability is the inclusion at the level of beliefs: $B_0 \in \mathrm{ERViab}_{\mathcal{I}}(\mathcal{V}) \Rightarrow B_0 \subseteq \mathrm{RViab}(\mathcal{V})$. Each inclusion can be strict, with a distinct cause: robust contraction arises from disturbances; epistemic contraction from indistinguishability; institutional contraction from restricted authority, enforcement, and allocation. The present paper characterizes the *epistemic* contraction: the mechanisms by which indistinguishability empties or shrinks the kernel.

2.2 The safe-control correspondence

For a constraint set $\mathcal{V}$ described by finitely many C^1 constraint functions, $\mathcal{V} = \{x : q_j(x) \geq 0,\ j = 1..m\}$, define the **safe-control correspondence** at $x \in \mathcal{V}$:

$$\mathcal{R}_{\mathcal{V}}(x) \;=\; \Big\{u \in U(x) : \forall d \in D(x),\ \forall j \text{ with } q_j(x) = 0,\ \nabla q_j(x)\cdot f(x,u,d) \geq 0\Big\},$$

the set of controls whose drift is inward (subtangential) on every active constraint face, uniformly over the disturbance. For x in the interior of $\mathcal{V}$ (no active constraints), $\mathcal{R}_{\mathcal{V}}(x) = U(x)$. Two levels of the tangency condition are distinguished throughout. (i) *Local reading.* The correspondence is applied to the constraint set itself: by Nagumo's theorem in its robust form, if $\mathcal{V}$ is closed and $\mathcal{R}_{\mathcal{V}}(x) \neq \varnothing$ on $\partial\mathcal{V}$ with a measurable selection $u(x) \in \mathcal{R}_{\mathcal{V}}(x)$, then $\mathcal{V}$ is itself robustly invariant, i.e. $\mathrm{RViab}(\mathcal{V}) = \mathcal{V}$ (Aubin, 1991; Frankowska, 1989). (ii) *Kernel reading.* When $\mathcal{V}$ is not invariant, viability of a point $x_0 \in \mathcal{V}$ requires instead a selection $u(x) \in \mathcal{R}_G(x)$ along the trajectory, where $G = \mathrm{RViab}(\mathcal{V})$ and $\mathcal{R}_G$ is the same correspondence computed on the kernel: points of $\mathcal{V} \setminus K$ may admit controls that are instantaneously safe while every trajectory must exit later. The obstruction theorems below are *local certificates against* $\mathcal{R}_{\mathcal{V}}$: exhibiting a state at which even the weaker pointwise condition fails certifies exclusion from the kernel, and this is the reading in which the certificates are used. The correspondence is the bridge between the pointwise (tangency) and the global (kernel) description, and it is the object the observation structure acts upon: an observation-based policy can select controls only as a function of the observation record.

2.3 Observation structures, information states, and epistemic kernels

Conventions. Throughout, the plant evolves in continuous time and a policy acts at review times, holding the selected action until the next review; the observation record is generated by the observation map O (the error, hidden-parameter, and delay extensions are noted below). A policy is observation-based if its action depends on the record only through the information set B_t; under the set-membership semantics the information set is a sufficient statistic, so observation-based and record-based policies coincide. The disturbance is chosen adversarially after the control (the Isaacs order of Theorems 1 and 2), an action outside $U(x)$ at a compatible state x is itself a failure (Section 2.4), and constraint violation is the strict inequality $q < 0$.

An **observation structure** $\mathcal{I} = (Y, O)$ is a measurable space Y and an observation map $O : X \to Y$. The policy observes $y(t) = O(x(t))$. At time t, the **information set** (belief) of a regulator who knows the dynamics, the disturbance class, and the record $(y(s))_{s \le t}$ is

$$B_t = \Big\{\xi(t) : \xi(0) \in B_0,\ \dot{\xi}(s) = f(\xi(s), u(s), d(s)),\ d(s) \in D(\xi(s)),\ O(\xi(s)) = y(s) \text{ for all observed } s \le t\Big\},$$

where $\xi(\cdot)$ ranges over candidate trajectories: the set of current states compatible with the record. We assume the standard set-membership semantics, so that B_t contains every state consistent with the observations and the applied controls. (With observation error, the equality becomes membership $y(s) \in H(\xi(s), v(s))$; with hidden parameters the belief lives in $X \times \Theta$; with delays it lives in a history space — the extensions are noted as open in Section 6.5.) A policy is **observation-based** if $u(t)$ depends on the record only through B_t; this is a special case of record-based policies, and the two coincide when the belief is a sufficient statistic for the decision problem, which holds in the set-membership semantics used throughout. We write $U^B(B) = \bigcap_{x \in B} U(x)$ for the controls available at every compatible state — the only controls an observation-based policy may select without risking inadmissibility.

Definition 1 (robust epistemic kernel)**.** $B_0 \in \mathrm{ERViab}_{\mathcal{I}}(\mathcal{V})$ if there exists a nonanticipative observation-based policy π such that, for every compatible initial state $x_0 \in B_0$ and every admissible disturbance realization $d(\cdot)$, the trajectory of $\dot{x} = f(x, u(t), d(t))$ generated from x_0 by π's actions under $d(\cdot)$ — the record being the one that $(\pi, d(\cdot))$ generates — remains in $\mathcal{V}$ for all time. (This is the standard exists-strategy-for-all-realizations form. The quantifier order is essential: the record is generated by the realization, so conditioning the admissible realizations on the record would be circular.) Admissible initial beliefs are the compact information sets $B_0 \subseteq X$ generated by the observation structure — the observation fibres $O^{-1}(y)$, $y \in O(X)$, in the constructions of Sections 3.2–3.4. The existential (non-robust) counterpart $\mathrm{EViab}_{\mathcal{I}}(\mathcal{V})$ is the contrast class defined below.

We use ERViab throughout, since the disturbance classes of sustainability problems are adverse by construction. The theorems are stated for this robust notion; they do not transfer a fortiori to the non-robust contrast class $\mathrm{EViab}_{\mathcal{I}}(\mathcal{V})$, in which there exist an observation-based policy and *some* admissible disturbance realization under which every compatible trajectory remains in $\mathcal{V}$ for all time. Since the robust kernel is contained in the non-robust one, an observation-based policy may exist there that no robust policy survives.

The thesis is now precise: **the epistemic kernel is the greatest recursively viable collection of information states in the sense of the estimation-space reduction cited in Section 5; common-action compatibility is a necessary one-step condition, but not by itself a complete characterization — and the obstruction calculus certifies, mechanism by mechanism, the information states outside it.**

2.4 Notation

We collect the principal symbols used throughout the paper and fix the two structural symbols: the **observation structure** is always $\mathcal{I}$ (calligraphic I) and the **institution** of Section 6.4 is always $\mathcal{J}$ (calligraphic J); no other letter serves either role. The state space is $X \subseteq \mathbb{R}^n$; the control set is U (set-valued $U(x)$ at state x); the disturbance set is D (set-valued $D(x)$). The dynamics are $\dot{x} = f(x, u, d)$. The closed constraint set is $\mathcal{V} \subseteq X$, described by finitely many C^1 constraint functions q_j ($j = 1, \ldots, m$). The safe-control correspondence at $x \in \mathcal{V}$ is $\mathcal{R}_{\mathcal{V}}(x)$. The observation structure is $\mathcal{I} = (Y, O)$ with observation map $O : X \to Y$; the information set (belief) at time t is B_t; the initial belief is B_0 — admissible initial beliefs are the compact information sets the observation structure generates, the observation fibres $O^{-1}(y)$, $y \in O(X)$, in the constructions. The common admissible action set at belief B is $U^B(B) = \bigcap_{x \in B} U(x)$ — the only controls an observation-based policy may select without risking inadmissibility, under the convention, used from Proposition 1 on, that an action outside $U(x)$ at a compatible state is itself failure; the common safe-action set is $\mathcal{R}^B_{\mathcal{V}}(B) = \bigcap_{x\in B}\mathcal{R}_{\mathcal{V}}(x)$. The tube-safe action set at review length Δ is $\mathcal{A}_{\mathrm{tube}}(B,\Delta) = \{a : \mathrm{Reach}_{[0,\Delta]}(B,a) \subseteq \mathcal{V}\}$, where $\mathrm{Reach}_{[0,\Delta]}(B,a)$ is the *robust* — all-disturbance — reachable set of the system from B under the held action a over $[0,\Delta]$. The first informative observation time is T_{obs}: the first time the observation record discriminates between compatible branches — before it, the branches of Theorem 3's (H4.1) produce identical records. The adverse-selection correspondences of the exit certificates are $D_\varepsilon(x,u)$ (Theorem 1) and $D_\eta(x)$ (Theorem 2's proof, local). The upper right Dini derivative of q in direction v is $D^+q(x;v) = \limsup_{h\downarrow 0}[q(x+hv)-q(x)]/h$; for q of class C^1 — the class assumed in the exit theorems — $D^+q(x;v) = \nabla q(x)\cdot v$.

The kernels of the hierarchy are $\mathrm{Viab}(\mathcal{V};U,\pi_{\mathrm{perf}})$ (perfect information), $\mathrm{RViab}(\mathcal{V})$ (robust), and $\mathrm{ERViab}_{\mathcal{I}}(\mathcal{V})$ (the robust epistemic kernel — Definition 1, the object of this paper), with $\mathrm{proj}_\exists(\mathfrak{B}) = \bigcup_{B\in\mathfrak{B}} B$ the existential projection of a collection $\mathfrak{B}$ of information states. The non-robust counterpart $\mathrm{EViab}_{\mathcal{I}}(\mathcal{V})$ is the non-robust contrast class (Section 2.3). The institutionally restricted kernel is $\mathrm{IRViab}_{\mathcal{J}}(\mathcal{V})$ (Section 6.4). The policy classes are π_{perf} (state feedback), Π_{output}, Π_{state}, and Π_{CE} — the certainty-equivalence class defined in Remark 3: fixed perfect-information regulation laws applied uncorrected to the observation. The policy-specific safety set of Section 4.1 is Safe^{Π}_{T}; the certainly-safe set of Corollary 1 is $\mathcal{Y}_{\mathrm{safe}}$.

The four correspondences of this section are one skeleton. An observation-based regulator selects a single response for an entire information set — a held action, a feedback segment, or a safe/unsafe verdict — and observation-based viability is the existence of a response that is simultaneously admissible, safe, and recursively viable for every compatible state and disturbance history. Each obstruction certificate of Sections 3 and 4 is a checkable sufficient condition under which one of these response correspondences is empty: $U^B(B)$ (admissibility), $\mathcal{R}^B_{\mathcal{V}}(B)$ (instantaneous safety), $\mathcal{A}_{\mathrm{tube}}(B,\Delta)$ (review-interval safety), or the recursive predecessor of Section 3.5. The mechanisms differ in which correspondence fails, and the obstructions are nested (Proposition 3).

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3. The Obstruction Calculus

3.1 The finite-time exit certificate

The base obstruction operates before any observation-theoretic argument. It is a drift condition under which *no* control — of any information structure — can be viable. In management terms: when the dynamics themselves force exit before any control can react, no monitoring design can rescue viability, and the certification problem is closed without any observation-theoretic reasoning.

Theorem 1 (finite-time exit certificate)**.** Let $q : X \to \mathbb{R}$ be a constraint function of class C^1 — the class assumed throughout the exit theorems — with $\mathcal{V} \subseteq \{q \ge 0\}$, and suppose the following hypotheses hold.

(H1.1) There exist constants $a > 0$ and $\varepsilon > 0$ such that on the strip $\mathcal{S}_a = \{x : 0 \le q(x) \le a\}$,

$$\sup_{u\in U(x)} \inf_{d\in D(x)} D^+q(x; f(x,u,d)) \le -\varepsilon \qquad \forall x \in \mathcal{S}_a, \tag{1}$$

where $D^+q(x;v) = \limsup_{h\downarrow 0}[q(x+hv)-q(x)]/h$ is the upper right Dini derivative of q in direction v (for C^1 q, $D^+q(x;v) = \nabla q(x)\cdot v$ — the form the proof uses).

(H1.2) **Closed-loop existence of the adverse realization.** The disturbance correspondence is lower semicontinuous, or constant, on the strip, so that the state-dependent adverse selection constructed in the proof is realized by an admissible control–disturbance pair for which the Carathéodory existence theorem applies. (Alternatively, without (H1.2), the certificate is read against the convexified (relaxed) inclusion $\dot{x} \in \operatorname{co} f(x, u(t), D_\varepsilon(x, u(t)))$, which admits solutions under Marchaud-type conditions and preserves the drift inequality because the constraint is convex in the velocity; closure and relaxation are the standard tools for this step.)

Then, under hypotheses (H1.1)–(H1.2), for every admissible control and every initial state $x_0 \in \mathcal{S}_a$ there exists an admissible disturbance realization under which the trajectory attains $q(x(t)) < 0$ — hence leaves $\{q \geq 0\} \supseteq \mathcal{V}$ — by every time strictly larger than $q(x_0)/\varepsilon$: $\inf\{t : q(x(t)) < 0\} \leq q(x_0)/\varepsilon$, in particular within time at most a/ε (since $q(x_0) \leq a$). (Convention, used in both exit theorems: *violation* is the strict inequality $q < 0$; at $q = 0$ the state may still lie on $\partial\mathcal{V} \subseteq \mathcal{V}$.)

Proof. Trajectories of $\dot{x} = f(x, u(t), d(t))$ exist in the Carathéodory sense under the standing hypotheses — D compact-valued, f measurable in t and continuous in (x, u, d), linearly bounded — for every measurable control–disturbance pair. The adverse realization selected below is *state-dependent*, so its closed-loop realization is an additional requirement — the content of (H1.2) (D lower semicontinuous or constant, or the convexified reading of its parenthetical); convexity of D was not needed for the closure step — closure and relaxation are the standard tools for it. Fix an admissible control $u(\cdot)$. For each $x \in \mathcal{S}_a$, condition (1) gives $\inf_{d \in D(x)} D^+q(x; f(x, u(t), d)) \leq -\varepsilon$. Define the adverse-selection correspondence

$$D_\varepsilon(x, u) \;=\; \big\{d \in D(x) : D^+q(x; f(x, u, d)) \leq -\varepsilon\big\},$$

which has nonempty values on $\mathcal{S}_a$ by (1). For q of class C^1 the Dini derivative reduces to $\nabla q(x) \cdot f(x, u, d)$, continuous in (x, u, d); with the closed graph and compact values of D (Section 2.1), D_ε then has closed graph and compact values, and the measurable-selection theorem (Aubin and Frankowska, 1990, Thm. 8.1.3) supplies a measurable selection $d(\cdot)$ with $d(t) \in D_\varepsilon(x(t), u(t))$ along the trajectory. (The proof is carried out for q of class C^1, as assumed; the locally Lipschitz case requires a separate comparison argument for almost-everywhere-differentiable arcs and is not treated.) Along the realization $(u(\cdot), d(\cdot))$ from x_0, the fundamental theorem of calculus along the absolutely continuous trajectory integrates the inequality to

$$q(x(t)) \;\leq\; q(x_0) - \varepsilon t$$

as long as $x(t)$ remains in $\mathcal{S}_a$. Hence $q \leq 0$ at a time not exceeding $q(x_0)/\varepsilon \leq a/\varepsilon$. If the trajectory has not already left the strip by time $t^* + \delta$ for some $\delta > 0$, the comparison continues to apply and gives $q(x(t^* + \delta)) \leq q(x_0) - \varepsilon(t^* + \delta) \leq -\varepsilon\delta < 0$ with $t^* = q(x_0)/\varepsilon$; in either case the trajectory violates $\{q \geq 0\} \supseteq \mathcal{V}$ before every time strictly larger than $q(x_0)/\varepsilon$. The selection $d(t) \in D_\varepsilon(x(t), u(t))$ is itself nonanticipative feedback of the pair (x, u) — a disturbance strategy in the Isaacs order, responding to the current state and control without anticipating future controls — which is what the certificate requires; no Elliott–Kalton upgrade to a closed-loop strategy is required for the stated claim. $\square$

Remark 1 (comparison-function form)**.** Hypothesis (H1.1) uses a constant drift margin ε. The certificate extends to state-dependent decline: if there is a continuous $\alpha : (0, a] \to (0, \infty)$ with

$$\sup_{u \in U(x)} \inf_{d \in D(x)} D^+q(x; f(x, u, d)) \leq -\alpha(q(x)) \qquad \forall x \in \mathcal{S}_a,$$

then the exit-time bound sharpens to

$$t_{\text{exit}} \leq \int_0^{q(x_0)} \frac{ds}{\alpha(s)},$$

whenever the integral is finite. The proof replaces the linear comparison $q(x(t)) \leq q(x_0) - \varepsilon t$ with the solution r of $\dot{r} = -\alpha(r)$, $r(0) = q(x_0)$, by the standard comparison argument. If $\alpha(0) = 0$ and the integral diverges, the conclusion weakens to asymptotic approach rather than finite-time exit; the constant-margin case $\alpha \equiv \varepsilon$ is the sharp finite-time instance.

Two readings of (1) matter. First, it is an **Isaacs-type condition**: the disturbance chooses after the control, so the certificate asserts that the disturbance can *enforce* exit against any policy. Second, (1) is an outward-drift certificate *dual in spirit, but not logically complementary*, to an inward barrier condition. The pointwise robust barrier reading is $\sup_u \inf_d D^+q \geq 0$ — some control keeps the drift inward against every disturbance — and the two conditions leave an unresolved intermediate region in which neither holds (Section 6.2). Theorem 1 is the "unsafety" side of that duality, and it is unconditional: it needs no observation argument because it defeats all controls, including clairvoyant ones.

3.2 Epistemic emptiness by admissibility: a minimal construction

The remaining obstructions apply to systems in which every state is individually viable under full information and the kernel empties only under the observation structure. The minimal construction of this section empties the kernel through *admissibility* — a state-dependent control set whose common action is forced and unsafe at the boundary; the common-action obstruction of the next section and Example 1's hidden-parameter conflict are the informational cases proper, merging states with incompatible safe controls. In ecological terms, this is the situation in which each stock, considered on its own, is manageable; the failure arises because the indicator cannot tell the manager which stock is present.

Proposition 1 (epistemic emptiness by admissibility — a minimal construction)**.** There exists a system with $\mathrm{Viab}(\mathcal{V}; U, \pi_{\mathrm{perf}}) = \mathcal{V} \neq \varnothing$ such that, for a non-injective observation map O, every admissible initial information state — the observation fibres $B_0 = O^{-1}(O(S_0))$, $S_0 \in \mathcal{V}$ — lies outside $\mathrm{ERViab}_{\mathcal{I}}(\mathcal{V})$: the epistemic kernel over the fibre-induced initial beliefs is empty. The mechanism is *admissibility* rather than a hidden mode: the state-dependent control set of the construction leaves a single common action at the merged belief — one that is outward-drifting at the boundary state — so the emptiness is the common-action obstruction (Theorem 2's safety case) at minimal size, produced by the control-set primitive rather than by information; the hidden-mode instance is Example 1.

Proof (by construction). Let $X = [1, 2]$, $\dot{S} = u - r(S)$ with $r : [1, 2] \to \mathbb{R}_{++}$ continuous and **injective on** $[1, 2]$ (e.g. $r(S) = S - 1 + c$, $c > 0$, strictly increasing), $U(S) = \{0, r(S)\}$, $\mathcal{V} = [1, 2]$, and the constant observation $O(S) \equiv 0$. Under perfect information, the state-feedback control $u = r(S)$ gives $\dot{S} = 0$, so every $S_0 \in \mathcal{V}$ is viable: $\mathrm{Viab}(\mathcal{V}; U, \pi_{\mathrm{perf}}) = \mathcal{V}$. Under the constant observation, the initial information state is the fibre $B_0 = [1, 2]$, and the set-membership semantics propagates it: at time t the belief is $B_t = \Phi_t(B_0) \cap \mathcal{V}$, the image of the initial set under the flow generated by the applied control — under a constant observation with no exit observation, this flow-image coincides with the record-compatibility definition of Section 2.3. The common control set at the initial belief is

$$U^B(B_0) = \bigcap_{S \in [1,2]} \{0, r(S)\} = \{0\},$$

because $r(S) > 0$ for every S and, by injectivity on $[1, 2]$, no single control equals $r(S)$ for all S. An observation-based policy may select time-varying actions, but robust admissibility requires the selected action to lie in $U^B(B_t)$ at every t; the propagated belief B_t is nonempty and, while the state remains in $[1, 2]$, $U^B(B_t) = \{0\}$ by the same computation (the flow of $\dot{S} = -r(S)$ with $r > 0$ moves B_t monotonically downward, preserving the injectivity of r on the surviving interval). Hence the policy must hold $u(t) = 0$ almost everywhere, giving $\dot{S} = -r(S) \leq -\min_{[1,2]} r < 0$; every compatible trajectory exits $\mathcal{V}$ downward in finite time. The same argument applies verbatim to any initial fibre, since O is constant. Hence $\mathrm{ERViab}_{\mathcal{I}}(\mathcal{V})$ contains no fibre-induced initial belief, while beliefs that are singletons (exact prior knowledge) are not admissible initial states of this observation structure. The system has no disturbance and no estimation error: the failure is caused entirely by the non-injectivity of O. $\square$

The construction isolates an admissibility mechanism: the observation merges states whose *admissible* controls differ — the merged belief admits a common admissible control (the singleton $\{0\}$) but no common *safe* control, because that singleton is outward-drifting at the boundary state $S = 1$. The "inadmissible action = failure" convention is used explicitly: an action outside $U(x)$ at a compatible state x is itself failure, and the state-dependent control set $U(S) = \{0, r(S)\}$ is a modelling primitive carrying that convention (Theorem 2's admissibility case uses the same convention, stated there). The example also shows the empty-kernel phenomenon at its minimal size — two control values, one scalar state — as the minimal instance of Theorem 2 rather than a mechanism of equal rank; with a constant control set $[0, \bar{u}]$ in the same plant the fibre is epistemically viable (a constant control $u = r(m)$, $m \in (1, 2)$, drives every state of the fibre to the rest point $m \in (1, 2)$), so the construction is an admissibility mechanism rather than a hidden-mode mechanism. The next theorem states the general obstruction, of which this construction is the minimal instance.

3.3 The instantaneous common-action obstruction

Theorem 2 (common-action obstruction)**.** Let $B \subseteq \mathcal{V}$ be an information set of the specified observation structure (the safe-control correspondence $\mathcal{R}_\mathcal{V}$ is defined on $\mathcal{V}$, so the belief is taken inside it). Suppose the following hypotheses hold.

(H3.1) The selected action is held until the next observation (sampled review: no informative observation arrives before the next action must be selected, and the policy holds the selected action over the review interval — the institutional reading; without a positive dwell time the instantaneous statement below must be replaced by the tube-safety statement below).

(H3.2) Either the **common admissible action set** is empty,

$$U^B(B) \;=\; \bigcap_{x \in B} U(x) \;=\; \varnothing, \qquad \text{(admissibility obstruction)}$$

or the **common safe-action set** is empty while common admissibility holds,

$$\mathcal{R}^B_\mathcal{V}(B) \;:=\; \bigcap_{x \in B} \mathcal{R}_\mathcal{V}(x) \;=\; \varnothing \quad \text{with} \quad U^B(B) \neq \varnothing, \qquad \text{(safety obstruction)} \tag{2}$$

(H3.3) **Closed-loop existence of the local adverse selection (safety case).** The disturbance correspondence is lower semicontinuous, or locally constant near the boundary point at which the obstruction fires, so that the local adverse selection of the proof is realized along the compatible trajectory; alternatively, the certificate is read against the convexified (relaxed) inclusion, which preserves the signed drift (the constraint is convex in the velocity).

Then, under hypotheses (H3.1)–(H3.3), $B \notin \mathrm{ERViab}_\mathcal{I}(\mathcal{V})$: every compatible state may be individually robustly viable, and the belief is nevertheless nonviable, because the state-specific safe actions are incompatible.

Proof. Let π be any observation-based policy, and let $a = \pi(B)$ be the action it selects at the information set B.

Admissibility case. If $a \notin U^B(B)$, then for some compatible state $x \in B$ the action is inadmissible ($a \notin U(x)$): at that state the selected action is not an admissible control, so the policy fails outright on the compatible branch from x — an obstruction independent of the safe set.

Safety case. Otherwise $a \in U^B(B)$. Because $\bigcap_{x\in B} \mathcal{R}_\mathcal{V}(x) = \varnothing$ while $\mathcal{R}_\mathcal{V}(x) = U(x)$ at interior points of $\mathcal{V}$ (no constraint is active there) and $a \in U(x)$ for every $x \in B$, the emptiness of the intersection must be caused by the boundary: there is a state $\bar{x} \in B \cap \partial\mathcal{V}$ with $a \notin \mathcal{R}_\mathcal{V}(\bar{x})$. By the defining inequality of the safe-control correspondence, there are an active constraint j (with $q_j(\bar{x}) = 0$) and a disturbance $\bar{d} \in D(\bar{x})$ such that

$$\nabla q_j(\bar{x}) \cdot f(\bar{x}, a, \bar{d}) < 0.$$

With $\eta = -\nabla q_j(\bar{x}) \cdot f(\bar{x}, a, \bar{d}) > 0$, define the adverse-selection correspondence $D_\eta(x) = \{d \in D(x) : \nabla q_j(x) \cdot f(x, a, d) \leq -\eta/2\}$ on a neighbourhood of $\bar{x}$; by continuity of the right-hand side in (x, d) and the closed graph of D (Section 2.1), it has nonempty compact values near $\bar{x}$ and admits a measurable local selection $d(x)$ with $d(\bar{x}) = \bar{d}$. Let the disturbance realize this selection on a small initial interval (its closed-loop realization is hypothesis (H3.3)'s content — D lower semicontinuous or locally constant near $\bar{x}$, or the convexified reading); then along the compatible trajectory from $\bar{x}$ (compatible, since $\bar{x} \in B$), q_j strictly decreases from $q_j(\bar{x}) = 0$ at rate at least $\eta/2$, so the trajectory leaves $\mathcal{V}$ within the first review period — before any informative observation can arrive, whichever action the policy takes. Since π was arbitrary, no observation-based policy keeps every compatible trajectory in $\mathcal{V}$: $B \notin \mathrm{ERViab}_{\mathcal{I}}(\mathcal{V})$. □

Condition (2) is checkable: it requires computing the intersection of the safe-control sets over the information set, which for polyhedral constraint and control sets is a finite linear program. When the intersection is nonempty, the common action is a candidate for viability — Theorem 2 is a *necessary* condition with a constructive witness of failure, not a sufficient condition for success. For polyhedral data the emptiness certificate is a Farkas pair: with safe actions given by finitely many affine inequalities $A_i u \leq b_i$ across compatible states, stacked as $Au \leq b$, the intersection is empty if and only if there exists $\mu \geq 0$ with $\mu^\top A = 0$ and $\mu^\top b < 0$ (Farkas, 1902; Gale, 1960) — a finite, independently checkable certificate of epistemic infeasibility, the control-space analogue of the material-substitution separation certificates in the assessment-separation analysis of Abaee (2026).

A worked certificate (two floors, one scalar action). Two floors demand incompatible extraction rates on a single control $u \in [0, 1]$. At the compatible state x_1 (floor 1 active) the safe controls are $u \leq 0.4$; at x_2 (floor 2 active) they are $u \geq 0.6$. The observation merges x_1 and x_2, so a common safe action must satisfy both. Writing the constraints as $Au \leq b$ with $A = (1, -1)^\top$ and $b = (0.4, -0.6)^\top$, the stacked system $u \leq 0.4$, $-u \leq -0.6$ is infeasible; the Farkas certificate is $\lambda = (1/2, 1/2)$ (normalized, $\|\lambda\|_1 = 1$), for which $\lambda^\top A = 0$ and $\lambda^\top b = -0.1 < 0$. Under the normalization $\|\lambda\|_1 = 1$, the magnitude $|\lambda^\top b| = 0.1$ is the infeasibility margin of the stacked system, shrinking to zero exactly as the two safe-action sets come to touch. The multiplier exhibits the contradiction as a convex combination of the two floor constraints: no common extraction rate respects both floors, and any observation that separates x_1 from x_2 restores one-step viability (Figure 1).

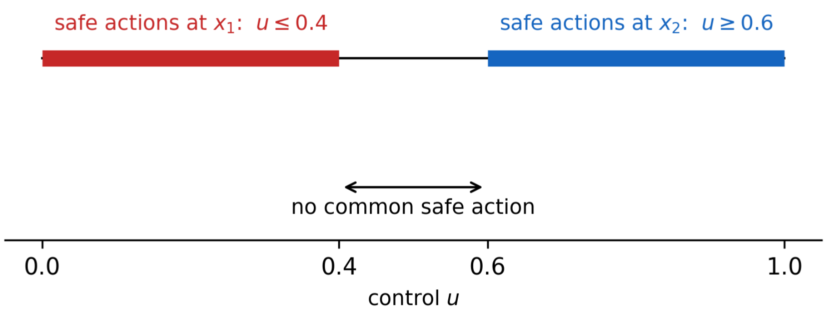

Figure 1: Incompatible safe controls. The states x_1 and x_2 share an observation, but their safe-action sets $[0, 0.4]$ and $[0.6, 1]$ are disjoint, so no common safe action exists (Theorem 2).

The tube-safety form. The instantaneous condition (2) is the $\Delta \downarrow 0$ boundary approximation of the exact one-step obstruction: with the action held over a review interval of length Δ, the exact condition is the emptiness of the tube-safe action set

$$\mathcal{A}_{\mathrm{tube}}(B, \Delta) \;=\; \Big\{ a : \mathrm{Reach}_{[0,\Delta]}(B, a) \subseteq \mathcal{V} \Big\},$$

with $\mathrm{Reach}_{[0,\Delta]}(B, a)$ the robust (all-disturbance) reachable set (the reading specified in Section

2.4), and $\mathcal{A}_{\rm tube}(B,\Delta) = \varnothing$ certifies $B \notin \mathrm{ERViab}_{\mathcal{I}}(\mathcal{V})$ for every review interval of length Δ and every target collection. The theorems above and below use the instantaneous and timing readings; the tube form is the unifying one-step object.

Proposition 2 (uniform margin gives a tube obstruction at every review length)**.** Let $B \subseteq \mathcal{V}$ be a compact information set with $U^B(B) \neq \varnothing$, f and ∇q_j continuous, and suppose there is $\eta > 0$ such that for every $a \in U^B(B)$ there exist a boundary state $x_a \in B \cap \partial\mathcal{V}$, an active constraint j with $q_j(x_a) = 0$, and a disturbance $d_a \in D(x_a)$ satisfying

$$\nabla q_j(x_a) \cdot f(x_a, a, d_a) \leq -\eta.$$

Under the closed-loop existence convention of Theorem 2, the tube-safe action set is then empty at every review length,

$$\mathcal{A}_{\rm tube}(B,\Delta) = \varnothing \qquad \forall\, \Delta > 0,$$

and hence $B \notin \mathrm{ERViab}_{\mathcal{I}}(\mathcal{V})$ under sampled review of any length.

Proof. Fix $a \in U^B(B)$ and the triple (x_a, j, d_a). By continuity of $(x,u,d) \mapsto \nabla q_j(x) \cdot f(x,u,d)$ and the closed graph of D, the drift inequality holds with margin $-\eta/2$ on a neighbourhood of (x_a, a, d_a), and a measurable local selection $d(x)$ realizes it. Along the trajectory from x_a under the held action a and this adverse selection, $q_j(x(t)) \leq q_j(x_a) - (\eta/2)\,t = -(\eta/2)\,t < 0$ for every small $t > 0$, so the trajectory leaves $\mathcal{V}$ in arbitrarily small positive time; hence $\mathrm{Reach}_{[0,\Delta]}(B,a) \nsubseteq \mathcal{V}$ for every $\Delta > 0$, i.e. $a \notin \mathcal{A}_{\rm tube}(B,\Delta)$. Since $a \in U^B(B)$ was arbitrary, $\mathcal{A}_{\rm tube}(B,\Delta) = \varnothing$ for every $\Delta > 0$, and the tube-safety form certifies $B \notin \mathrm{ERViab}_{\mathcal{I}}(\mathcal{V})$. □

Proposition 3 (obstruction ladder)**.** Under the standing regularity assumptions and the closed-loop existence convention of Theorem 2, the common response sets of Section 2.4 are nested:

$$\mathcal{A}_{\rm tube}(B,\Delta) \;\subseteq\; \mathcal{R}^B_{\mathcal{V}}(B) \;\subseteq\; U^B(B) \qquad \forall\, \Delta > 0,$$

and emptiness descends the ladder:

$$U^B(B) = \varnothing \implies \mathcal{R}^B_{\mathcal{V}}(B) = \varnothing \implies \mathcal{A}_{\rm tube}(B,\Delta) = \varnothing \implies B \notin \mathrm{ERViab}_{\mathcal{I}}(\mathcal{V}).$$

Proof. The second inclusion is by definition, since $\mathcal{R}_{\mathcal{V}}(x) \subseteq U(x)$ for every x. For the first, fix $a \in \mathcal{A}_{\rm tube}(B,\Delta)$, $x \in B$ with an active constraint j ($q_j(x) = 0$), and $d \in D(x)$. If $\nabla q_j(x) \cdot f(x,a,d) < 0$, then along the trajectory from x under the held action a and the constant disturbance d, $q_j(x(t)) = \nabla q_j(x) \cdot f(x,a,d)\,t + o(t) < 0$ for all small $t > 0$, so the trajectory leaves $\mathcal{V}$ in arbitrarily small time, contradicting $a \in \mathcal{A}_{\rm tube}(B,\Delta)$. Hence $\nabla q_j(x) \cdot f(x,a,d) \geq 0$ for every $d \in D(x)$ and every active j, i.e. $a \in \mathcal{R}_{\mathcal{V}}(x)$ for every $x \in B$, so $a \in \mathcal{R}^B_{\mathcal{V}}(B)$. □

The obstruction ladder. Each certificate is a sufficient condition for emptiness at one rung:

- *Admissibility obstruction* — $U^B(B) = \varnothing$: no single action is admissible at every compatible state (the base case of Proposition 1).
- *Common safe-action obstruction* — $\mathcal{R}^B_{\mathcal{V}}(B) = \varnothing$ with $U^B(B) \neq \varnothing$ (the safety case of Theorem 2).
- *Tube obstruction* — $\mathcal{A}_{\rm tube}(B,\Delta) = \varnothing$ (the tube-safety form).
- *Recursive obstruction* — in the finite-horizon setting the witness action sets of Section 3.5 complete the ladder $\mathcal{A}_N(B) \subseteq \mathcal{A}_{\rm tube}(B,\Delta) \subseteq \mathcal{R}^B_{\mathcal{V}}(B) \subseteq U^B(B)$.

The rungs are strict in general: $\mathcal{R}^B_{\mathcal{V}}(B) \neq \varnothing$ does not imply $\mathcal{A}_{\rm tube}(B,\Delta) \neq \varnothing$ for $\Delta > 0$, since an instantaneously tangent action may still exit in finite time — the gap that Proposition 2 closes under a uniform margin. The dynamic exit certificate of Theorem 1 sits below the ladder: it empties the response set at some compatible state even under full information, which is the strongest form of obstruction and needs no observation-theoretic argument.

Example 1 (hidden-mode conflict)**.** Let an unobserved parameter satisfy $\theta \in \{-1, +1\}$, with $\dot{z} = \theta u$, $u \in \{-1, +1\}$, and constraint $z \geq 0$. At $z = 0$: if $\theta = +1$, only $u = +1$ is safe; if $\theta = -1$, only $u = -1$ is safe. Both compatible states are individually robustly viable under full information (each admits its safe control), but the information set $B = \{(0, +1), (0, -1)\}$ satisfies $\mathcal{R}^B_{\mathcal{V}}(B) = \{+1\} \cap \{-1\} = \varnothing$, so $B \notin \mathrm{ERViab}_{\mathcal{I}}(\mathcal{V})$. This is a purely informational failure: no stochasticity and no estimation-quality argument is involved. The example is the two-action skeleton of every "the stock is either recovering or collapsing, and the safe policy differs between the two" situation.

3.4 The delayed-information obstruction

Theorem 3 (delayed-information obstruction)**.** Let B_0 be an initial information set. Suppose the following hypotheses hold.

(H4.1) No informative observation arrives before time $T_{\mathrm{obs}} > 0$ — formally, there are branches that are **observation-equivalent on** $[0, T_{\mathrm{obs}})$: they produce identical observation records throughout that interval, so the policy cannot distinguish them before T_{obs}; between observations the information set evolves by the set-membership semantics of Section 2.3.

(H4.2) There exist a constraint function q of class C^1 (the class of Theorem 1's proof) with $\mathcal{V} \subseteq \{q \geq 0\}$ and a constant $\varepsilon > 0$ such that, **for every open-loop control on the blind window** — every measurable $u(\cdot) : [0, T_{\mathrm{obs}}) \to \bigcup_{x \in B_0} U(x)$, the class of controls that any observation-based policy realizes before the first informative observation, where its actions are a fixed function of time — there are a compatible initial state $x^* \in B_0$ attaining $q(x^*) = \inf_{x \in B_0} q(x)$ (with B_0 compact and q lower semicontinuous; otherwise replace the infimum by $\inf q + \delta$ for an arbitrary δ-minimizer, let $\delta \downarrow 0$, and read the timing bound with $\inf q + \delta$) and an admissible disturbance realization such that, along the realized trajectory from x^* under that open-loop control, the record remains compatible and

$$D^+ q(x(t); f(x(t), u(t), d(t))) \;\leq\; -\varepsilon \qquad \text{while } q(x(t)) > 0, \tag{3}$$

(H4.3) The timing bound holds:

$$T_{\mathrm{obs}} \;>\; \frac{\inf_{x \in B_0} q(x)}{\varepsilon}. \tag{4}$$

Then, under hypotheses (H4.1)–(H4.3), $B_0 \notin \mathrm{ERViab}_{\mathcal{I}}(\mathcal{V})$: information may be accurate but arrive too late.

Proof. Fix any observation-based policy. By (H4.1) its actions on $[0, T_{\mathrm{obs}})$ are a fixed open-loop function of time — the record carries no informative observation before T_{obs}, so every observation-equivalent branch induces the same actions — and (H4.2), applied to that open-loop control, supplies a compatible initial state x^* with $q(x^*) = \inf_{x \in B_0} q(x)$ and an admissible disturbance realization satisfying (3) along the trajectory from x^* under those actions, observation-equivalent on $[0, T_{\mathrm{obs}})$ to the other compatible branches. For q of class C^1 — the class assumed in (H4.2) — the fundamental theorem of calculus along the absolutely continuous trajectory integrates (3) to

$$q(x(t)) \;\leq\; q(x^*) - \varepsilon t \;=\; \inf_{x \in B_0} q(x) - \varepsilon t$$

while $q(x(t)) > 0$, so the trajectory attains the violation convention of Theorem 1 — the strict inequality $q(x(t)) < 0$ — by every time strictly larger than $t^* = \inf_{x \in B_0} q(x)/\varepsilon$ (in particular $\inf\{t : q(x(t)) < 0\} \leq t^*$; the same argument with the δ-minimizer bounds the violation time by $(\inf q + \delta)/\varepsilon$), and t^* by (4) is strictly less than T_{obs} — the violation is strict and precedes the first informative observation. Until t^*, the record contains no informative observation: the violating branch is observation-equivalent to every other compatible branch before T_{obs}, so the policy cannot have altered its action in response to the violation before it occurs. Since the policy was arbitrary and the violating realization is compatible with the observation record, no observation-based policy is robustly viable: $B_0 \notin \mathrm{ERViab}_{\mathcal{I}}(\mathcal{V})$. □

The hypothesis (3) is the set-membership form of the drift condition, quantified over open-loop controls on the blind window — a condition on the data of the problem, not on the policies: for every such control there is a compatible state and disturbance enforcing the drift; the adversary selects the true state and the disturbance, and the observations cannot expose the selection in time. It is this open-loop form that gives the timing bound (4) its force — (H4.1) converts the policy's actions into one open-loop control, (H4.2) supplies the drift against it, and (4) then makes the obstruction epistemic rather than dynamic. The hypothesis is checkable in its special cases (constant observations; hold-until-T_{obs} policies) and is a template to be instantiated in the general case. Condition (4) is the **timing bound**: it states that the observation arrives *after* the enforced exit time — the cause of the obstruction; inverted, it reads as the design requirement that the first informative observation precede the enforced exit time $\inf q/\varepsilon$. Theorem 3 refines Theorem 2 quantitatively: the common-action obstruction is a **zero-margin formal analogue** of the delayed tube-safety obstruction. The two certificates are separate; the common-action obstruction is the zero-margin instance ($\inf_{x \in B} q(x) = 0$ with $T_{\mathrm{obs}} > 0$ — any strictly positive observation delay is then too late under a uniform outward drift), and the pair is not related as a $T_{\mathrm{obs}} \to \infty$ limit. It is the obstruction that governs discrete-review governance: a review interval longer than the worst-case exit time is an information structure in which the exit is inevitable.

Remark 2 (threshold form of the timing obstruction)**.** The delayed-information obstruction has a sharp threshold. Let $\sigma^*(B_0)$ be the guaranteed blind-window survival time: the largest duration such that some pre-observation policy keeps every observation-equivalent branch in $\mathcal{V}$ until then,

$$\sigma^*(B_0) \;=\; \sup_{u(\cdot)} \inf_{x_0 \in B_0,\, d(\cdot)} \tau(x_0, u, d),$$

with the infimum over branches that remain observation-equivalent on $[0, T_{\mathrm{obs}})$ and τ the first violation time under the convention of Theorem 3. Then

$$\sigma^*(B_0) < T_{\mathrm{obs}} \implies B_0 \notin \mathrm{ERViab}_{\mathcal{I}}(\mathcal{V}),$$

because before T_{obs} a policy cannot condition its action on which branch realizes. Condition (4) certifies this threshold: under (H4.1)–(H4.2) every branch exits by time $\inf_{x \in B_0} q(x)/\varepsilon$, hence $\sigma^*(B_0) \le \inf_{x \in B_0} q(x)/\varepsilon$, and (4) states that this upper bound lies below T_{obs}; the threshold itself may be strictly smaller, in which case nonviability holds even when (4) fails. The threshold is the blind-window response correspondence of the timing layer: its emptiness is the timing certificate, the delayed-information analogue of the tube and common-action correspondences.

3.5 Finite-horizon completeness (finite systems)

The certificates of Sections 3.1–3.4 are sufficient conditions, not a complete characterization of the epistemic kernel. In the finite setting the calculus is complete: backward recursion over beliefs is sound and complete on any finite horizon, and nonviability is certified by a finite obstruction tree.

Fix finite state, action, disturbance, and observation spaces X, A, D, Y, a transition map $x^+ = F(x, a, d)$, an observation map $y = O(x, a, d, x^+)$, a safe set $\mathcal{V} \subseteq X$, and admissible-action sets $U(x) \subseteq A$. For a belief $B \subseteq X$, an action a, and an observation y, the post-belief is

$$\mathrm{Post}(B, a, y) = \left\{x^+ \in X : \exists\, x \in B,\ d \in D(x),\ x^+ = F(x, a, d),\ y = O(x, a, d, x^+)\right\},$$

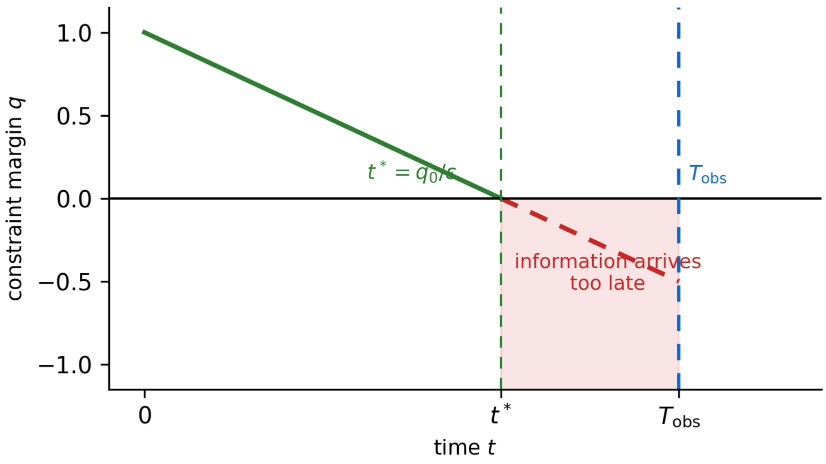

Figure 2: Delayed information. The constraint margin q reaches zero at $t^* = q_0/\varepsilon$, before the first informative observation at T_{obs}: the information is accurate but arrives too late (Theorem 3).

and y is possible under (B, a) if $\text{Post}(B, a, y) \neq \varnothing$. With $U^B(B) = \bigcap_{x \in B} U(x)$, define the one-step predecessor of a collection $\mathcal{C}$ of beliefs by

$$\text{Pre}(\mathcal{C}) = \left\{ B \subseteq \mathcal{V} : \exists\, a \in U^B(B) \text{ such that } \text{Post}(B, a, y) \in \mathcal{C} \text{ for every possible } y \right\},$$

and the finite-horizon kernels by $\mathcal{W}_0 = \{B : B \subseteq \mathcal{V}\}$, $\mathcal{W}_{k+1} = \text{Pre}(\mathcal{W}_k)$.

Theorem 4 (finite-horizon soundness and completeness)**.** An initial belief B_0 admits an observation-based policy that keeps every compatible trajectory in $\mathcal{V}$ for N steps if and only if $B_0 \in \mathcal{W}_N$.

Proof. ($\Rightarrow$) Induction on N. The case $N = 0$ is $B_0 \subseteq \mathcal{V}$, i.e. $B_0 \in \mathcal{W}_0$. If a policy is safe for N steps, its first action a is admissible at every compatible state, so $a \in U^B(B_0)$; for every possible observation y, the successor belief $\text{Post}(B_0, a, y)$ is safe for $N - 1$ steps under the same policy, so by induction $\text{Post}(B_0, a, y) \in \mathcal{W}_{N-1}$; hence $B_0 \in \text{Pre}(\mathcal{W}_{N-1}) = \mathcal{W}_N$.

($\Leftarrow$) Backward induction. If $B_k \in \mathcal{W}_{N-k} = \text{Pre}(\mathcal{W}_{N-k-1})$, choose the witness action $a_k \in U^B(B_k)$; after the observation y_k the successor belief $B_{k+1} = \text{Post}(B_k, a_k, y_k) \in \mathcal{W}_{N-k-1}$, and the recursion continues. By construction every reachable state over N steps lies in $\mathcal{V}$. $\square$

If $B_0 \notin \mathcal{W}_N$, the failure is certified by a finite **obstruction tree**: at each node (B, k) with $k \geq 1$, for every $a \in U^B(B)$ the adversary can choose a possible observation y with $\text{Post}(B, a, y) \notin \mathcal{W}_{k-1}$; the branches are the compatible observation-and-disturbance choices that force violation within N steps. This is the exact finite-horizon counterpart of the sufficient certificates of Sections 3.1–3.4.

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4. Certification Limits

4.1 Exact certification and the fibre criterion

The obstruction theorems of Section 3 concern *policies*. Sustainability governance also uses *certificates*: verdicts — safe or unsafe — computed from observations alone, without reference to a policy. The question is when such verdicts can be exact. For a manager operating under partial ecological information, this is the question of whether an indicator reading alone can ever settle that the system is, or is not, on the safe side of a specified floor.

Definition 2 (exact certifier)**.** For an admissible domain $Z \subseteq X$ and a safe set $K \subseteq Z$, an **exact certifier** based on the observation map $O : Z \to Y$ is a function $C : Y \to \{0, 1\}$ such that

$$C(O(z)) = 1 \iff z \in K \qquad \text{for every } z \in Z.$$

Proposition 4 (observation-fibre criterion)**.** An exact certifier based on O exists if and only if membership in K is constant on every observation fibre:

$$O(z_1) = O(z_2) \implies \big[z_1 \in K \iff z_2 \in K\big] \qquad \forall z_1, z_2 \in Z, \tag{5}$$

equivalently, $K = O^{-1}(O(K))$ on the admissible domain.

Proof. ($\Rightarrow$) If C exists and $O(z_1) = O(z_2)$, then $\mathbf{1}_K(z_1) = C(O(z_1)) = C(O(z_2)) = \mathbf{1}_K(z_2)$, so membership is fibre-constant.

($\Leftarrow$) If (5) holds, define $C(y) = \mathbf{1}_K(z)$ for any $z \in Z$ with $O(z) = y$; (5) makes C well defined on $O(Z)$ and exact by construction; extend C arbitrarily on $Y \setminus O(Z)$. A *measurable* exact certifier exists under the same condition on standard Borel spaces when $O(K)$ is measurable, taking $C = \mathbf{1}_{O(K)}$; measurability is the only additional requirement. □

The fibre criterion concerns static classification — whether membership in a safe set factors through the observation — and applies verbatim to three distinct certification problems, which should not be conflated: constraint certification ($K = \mathcal{V}$), robust viability certification ($K = \mathrm{RViab}(\mathcal{V})$), and policy-specific safety certification ($K = \mathrm{Safe}_T^\Pi$). The same theorem governs all three, but its governance meaning differs. The criterion is the zero-step, label-valued analogue of the common-action test: an observation-based verdict assigns a single label to an entire fibre, exactly as an observation-based action assigns a single action to an entire belief, so both are selector problems. They remain distinct — a fibre may fail exact certification while a policy is still viable, and exact static certification does not imply dynamic viability — which is why certification is treated as a layer separate from viability throughout.

Corollary 1 (safety-crossing fibres and the certainly-safe set)**.** If two admissible states share an observation and lie on opposite sides of the *same* component safety constraint — with the safe set of that certification problem read as the single floor $K = \{q_j \geq 0\}$; or, in the general form of the hypothesis, one of the two states lies in K and the other outside K — no exact observation-only certificate exists. (Crossing one floor of an intersection $K = \bigcap_j \{q_j \geq 0\}$ does not by itself put the pair on opposite sides of the intersection — the state satisfying floor j may violate floor i — hence the single-floor scoping.) The largest set of observations that can soundly be labeled safe without further information is the certainly-safe set

$$\mathcal{Y}_{\text{safe}} \;=\; \big\{y \in O(Z) : O^{-1}(y) \subseteq K\big\}.$$

Proof. In the general form (one state in K, one outside), the fibre violates (5) for that K, so Proposition 4 denies the certifier; in the single-floor form it violates (5) for the floor's own certification problem ($K = \{q_j \geq 0\}$). On $\mathcal{Y}_{\text{safe}}$ every compatible state is safe, so the constant verdict "safe" is sound; outside $\mathcal{Y}_{\text{safe}}$ some compatible state is unsafe, so no sound "safe" verdict exists there. □

The fibre criterion has a direct governance reading. A composite sustainability index is an observation map from the state of a socio-ecological system to a scalar; a floor is a safe set. Corollary 1 then states: an index whose fibres cross a floor boundary — whose value cannot distinguish a state violating the floor from one satisfying it — cannot support an exact certification of that floor, however carefully the index is thresholded. The structural requirement is not that each floor be measured separately, but that the observation *separate every admissible pair lying on opposite sides of a specified floor*; direct per-floor measurement is one sufficient design, not the unique one, and any statistic refining the floor's safe/unsafe partition certifies equally well. The certainly-safe set is the sound relaxation: it is the region where the index may certify safety; outside it, the index must fall silent.

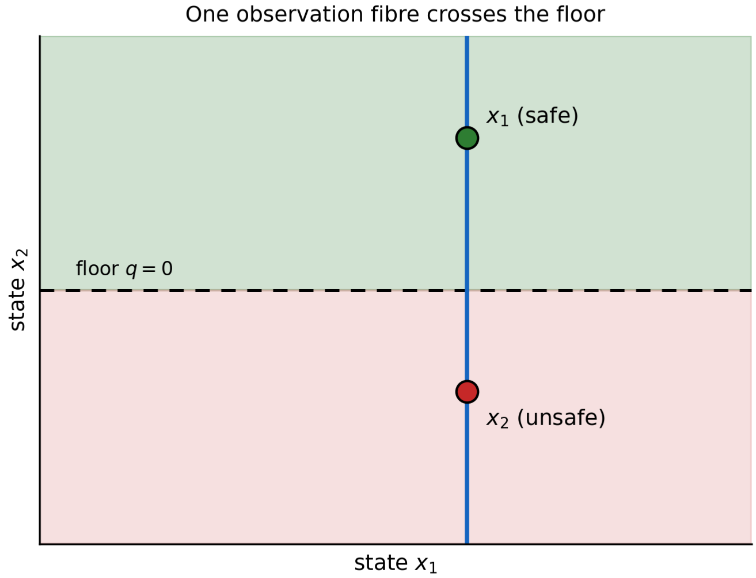

Figure 3: An observation fibre crossing a floor boundary. The observation $O(x) = x_1$ cannot separate the safe state $(0.6, 0.8)$ from the unsafe state $(0.6, 0.3)$ on the same fibre, so no exact observation-only certifier exists (Proposition 4).

4.2 The certainty-equivalence trap

The obstruction of Proposition 1 relies on a non-injective observation — information lost. The same emptying occurs with a fully *injective* observation when the policy class is restricted to certainty-equivalence controllers.

Remark 3 (certainty-equivalence obstruction)**.** Consider $\dot{S} = u - g(S)$ with $u \in [0, \bar{u}]$, g strictly increasing on $[S_{\min}, S^* + b]$, $g(0) = 0$, and $\mathcal{V} = [S_{\min}, S^*]$. Assume the range condition $g(S + b) \leq \bar{u}$ for every $S \in [S_{\min}, S^*]$, so that the certainty-equivalence command below is admissible rather than inadmissible. Under perfect information the feedback $u(t) = g(S(t))$ gives $\dot{S} = 0$, so every $S_0 \in \mathcal{V}$ is viable and $\mathrm{Viab}(\mathcal{V}; U, \pi_{\mathrm{perf}}) = \mathcal{V} \neq \varnothing$. Now take the injective, biased observation $\hat{S} = S + b$ with $b > 0$, and restrict the policy class to the **certainty-equivalence class** Π_{CE}, defined as the class of causal controllers obtained by taking a fixed *perfect-information regulation law* k of the plant and applying it directly to the uncorrected observation, $u = k(\hat{S})$ — no use of the observation map's structure (here, the bias b) enters the law. The trap below is the member $k = g$: $u = g(\hat{S})$. Then

$$\dot{S} = g(S + b) - g(S) > 0 \qquad \forall S,$$

and since g is strictly increasing on the compact interval $[S_{\min}, S^* + b]$, $\dot{S}$ is bounded below by a positive constant there; hence S strictly increases and exits above S^* in finite time from every $S_0 \in \mathcal{V}$:

$$\mathrm{Viab}(\mathcal{V}; U, \Pi_{\mathrm{CE}}) = \varnothing.$$

The mechanism is a *restriction of the policy class*, not a loss of information: for a known invertible observation the output-feedback class is as powerful as the state-feedback class, $\Pi_{\mathrm{CE}} \subsetneq \Pi_{\mathrm{output}} \cong \Pi_{\mathrm{state}}$, so $\mathrm{Viab}(\mathcal{V}; U, \Pi_{\mathrm{CE}}) \subsetneq \mathrm{Viab}(\mathcal{V}; U, \Pi_{\mathrm{output}}) = \mathrm{Viab}(\mathcal{V}; U, \Pi_{\mathrm{state}})$ — the kernel empties exactly when the controller does not apply the correction. An observer who inverts the bias, $u = g(\hat{S} - b) = g(S)$, recovers the perfect-information kernel — and lies outside Π_{CE}, since the composite law $g(\cdot - b)$ is not a perfect-information law of the plant applied uncorrected. The remark is the mechanism behind the monitoring-design consequence of Section 6.4: monitoring bounds observation error so that a desired state-feedback law is implementable, but the bound is useful only if the controller uses the correction; an uncorrected biased indicator empties the kernel that a corrected one preserves. The phenomenon is the viability-theoretic analogue of Witsenhausen's counterexample in stochastic control (Witsenhausen, 1968): the information pattern — not the dynamics and not the constraint — defeats the restricted controller class.

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5. The Sufficiency Landscape

For completeness and contrast, we record the sufficiency results against which the obstruction calculus is defined. The following results are standard; proofs appear in the cited sources.

(a) Veliov's output-feedback condition. Veliov (1993) considers the same problem as Section 3 (a tube in the state space under incomplete and inexact measurement) and gives a sufficient condition for the existence of an *output-feedback regulation map*: a set-valued feedback of the measurement under which all trajectories starting from the graph of the tube remain in it. Under perfect measurement his condition reduces to the classical viability condition (Haddad, 1981). The complementarity with this paper is exact: Veliov's theorem certifies existence; the obstruction calculus certifies nonexistence; a problem that satisfies neither requires a stronger observer or a finer observation structure.

(b) The estimation-tube programme. Cardaliaguet, Quincampoix, and Saint-Pierre (2007) pass from imperfect information in the measurement space to perfect information in an estimation space of *sets* of compatible states: the value functions of the two problems coincide, and the estimation-space problem admits a Dini-derivative characterization. The reduction is exact for value functions; what it does not supply is a *common prescription*. The exact relationship is stated in Section 6.3: carried out under the set-membership semantics, the estimation-space solution propagates the belief under a single control signal, so its controls *are* common controls and joint admissibility is built into the estimation-space problem's own admissibility. Pointwise selections taken without that joint-admissibility check need not be jointly admissible; the common-action obstruction (Theorem 2) is the certificate that no jointly admissible selection exists at the information state in question.

(c) Observer-and-buffer transfer. When a full-information feedback exists with a strict inward margin, and an observer supplies exponentially convergent estimates, output feedback of the estimate preserves safety on a buffered subset: if K_ζ is a compact controlled-invariant subset of the interior of the kernel, the state feedback $u = k(x)$ is Lipschitz with constant L_k, the observer satisfies $\|\hat{x}(t) - x(t)\| \leq Me^{-\lambda t}\|\hat{x}(0) - x(0)\|$, and the initial estimation error is small enough that $L_k M\|\hat{x}(0) - x(0)\|$ is absorbed by the margin, then K_ζ is viable under output feedback (observer-to-viability transfer with safety buffer; standard arguments in the observer-based control literature). Similarly, eroded kernels: if K is robustly invariant under full-state feedback with a strict inward margin on ∂K, and estimation and implementation errors are bounded by ζ, then an eroded set $K^{-c\zeta}$ is invariant under output feedback for a sensitivity constant $c > 0$ (erosion absorbs the error; see the buffer constructions in the literature cited in Section 1.3). These results bound the sufficiency side: under margins and convergence, output feedback *can* work; the obstruction calculus of Sections 3–4 states the conditions under which it *cannot*.

(d) The linear substitution alternative. For the finite linear model in which a resource-typed system must meet a demand vector through specified substitution pathways, exactly one of the following holds (Farkas, 1902; Gale, 1960): (i) there exists a non-negative pathway vector $a \geq 0$ satisfying the linear substitution constraints; or (ii) there exist multipliers $\alpha, \beta, \gamma \geq 0$ such that

$$\alpha^\top R + \beta^\top E - \gamma^\top Q \geq 0 \quad \text{componentwise}, \qquad \gamma^\top s^{\mathrm{req}} > \alpha^\top x + \beta^\top e.$$

The second statement is a certificate that the specified substitution pathways cannot meet demand within the typed resource and capacity bounds; writing all constraints as $Aa \leq \rho$, the pair is the Farkas lemma alternative. The multipliers are a separation certificate, not universal exchange rates: nonlinear, nonconvex, path-dependent, spatial, or irreversible technologies require their own feasibility analysis, and an elasticity fitted near one operating point cannot establish global substitutability. This is the feasibility-side complement of the observation-side obstructions of Sections 3–4: where the present paper's certificates bound what *information* can achieve, the substitution alternative bounds what *material pathways* can achieve, and the same infeasibility logic — exhibit the separating multiplier rather than assert impossibility — applies to both.

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6. Discussion

6.1 The shape of the calculus

The six mechanisms form a nonexhaustive taxonomy of information-theoretic failure. The exit certificate (Theorem 1) is the dynamic obstruction: it defeats all controls, so it needs no observation argument. The epistemic-emptiness construction (Proposition 1) exhibits the mechanism at minimal size — one scalar state, two admissible controls, one constant observation — in the admissibility form of Section 3.2. The common-action obstruction (Theorem 2) is the static obstruction: it defeats all policies at a single information state. The delayed-information obstruction (Theorem 3) interpolates between them in time. The fibre criterion (Proposition 4) is the certification obstruction: it limits not policies but verdicts. The certainty-equivalence trap (Remark 3) limits policy classes. Finite-horizon completeness (Theorem 4) supplies the exact finite-setting counterpart of these sufficient certificates.

A governance failure of observation can be diagnosed by which certificate applies. If the exit certificate holds, no monitoring design helps — the dynamics are adverse under every control. If only the common-action obstruction holds, the remedy is a finer observation structure (an informative measurement distinguishing the incompatible states). If the delayed-information obstruction holds with a slack timing bound, the remedy is a shorter review interval or a faster indicator. If the fibre criterion fails, the remedy is an observation that separates the sides of the floor rather than a better threshold on the aggregate.

6.2 Relation to barrier certificates

Barrier certificates certify safety of continuous and hybrid systems from a scalar function whose zero level set separates the unsafe region from all trajectories (Prajna and Jadbabaie, 2004; Prajna, Jadbabaie, and Pappas, 2007). The converse direction — that safety implies, under convex-duality conditions on density functions, the existence of a barrier certificate — is Prajna and Rantzer (2005); necessary-and-sufficient barrier characterizations for hybrid inclusions under mild regularity are Maghenem and Sanfelice (2019). Theorem 1 is the unsafety complement of that converse, not its observation-theoretic counterpart: it is a certificate of *unsafety* needing no observation structure — a Dini-drift condition under which failure is inevitable against all controls, observation-based or not — and it is constructive, exhibiting the enforcing disturbance and the exit time.

The two literatures meet at the boundary. When a converse barrier theorem applies, the outward-drift certificate can be compared with the failure of the corresponding inward condition; outside converse settings, the absence of a found barrier has no diagnostic force. The remaining certificates (Proposition 1, Theorem 2, Theorem 3, and Proposition 4) act at the level of the *information set itself*: state-space barrier constructions presuppose that the state — or a specified state estimate — is available for feedback, while observer-based and belief-space barrier constructions have been proposed that certify safety under a specified controller; neither addresses the emptiness of the common safe-action set at a given information state, which is the object of the common-action and timing obstructions.

6.3 Relation to estimation tubes

The estimation-tube programme shows that imperfect measurement can be absorbed into set-valued dynamics on an estimation space with no loss in value (Cardaliaguet, Quincampoix, and Saint-Pierre, 2007). The present theorems do not contradict that reduction; they delimit its reach. The reduction preserves *values* — whether the problem has a solution at a given information state — and its belief-state controls are common controls applicable to all compatible states under the specified semantics.

The obstruction calculus is not outside the estimation-space programme; it is the local certificate layer of it. The common-action obstruction is a finite certificate that the belief-state predecessor is empty, the timing bound quantifies when the predecessor fails for reasons of information arrival, and the fibre criterion states when no observation-based verdict can be exact at all. The two directions are thus complementary at the level of governance: belief-space viability answers *whether*, the obstruction calculus answers *why not, and what to change*.

6.4 Consequences for monitoring and indicator design

Five consequences follow directly from the certificates. Each translates a formal impossibility into a concrete design specification for monitoring systems under partial ecological information.

Timing. Theorem 3 gives the minimal monitoring frequency in closed form: the first informative observation must precede the enforced exit time $\inf q/\varepsilon$. A review interval longer than the worst-case time from the most-constrained compatible state — the minimum-q state, which has the least constraint slack — to constraint violation is an information structure under which the violation is undetectable in time, whatever the review then concludes. The bound is computable from the drift certificate (3), which a robustness analysis computes in any case.

Coarseness. Theorem 2 states that an observation structure merging two compatible states with incompatible safe controls is nonviable *at that information state*, even though both states are viable in isolation. The design consequence is that the observation must separate safety classes, not states: an indicator is exposed to Theorem 2's obstruction when some observation fibre crosses the safe-control partition of the state space — merging states whose safe controls differ is what the theorem certifies, and keeping every fibre within a single class is what removes that exposure; refinement is never harmful (a finer observation merges nothing the coarser one did not), though Theorem 2 claims no benefit beyond the removal, and other mechanisms may still certify nonviability on a finer partition.

Aggregation. Proposition 4 and Corollary 1 apply verbatim to composite indices, which are observation maps from the system state to a scalar. An index whose fibres cross a floor boundary cannot support exact floor certification; the certainly-safe set is the domain in which an index-based verdict is sound. Where separately-binding floors matter, exact certification requires an observation that separates every admissible pair on opposite sides of a floor (per-floor measurement is one sufficient design), a formal complement to the thesis, argued informally since Martinez-Alier, Munda, and O'Neill (1998), that weakly comparable values cannot be fused into one metric without loss. The requirement is sharpest where the floor constrains the productive base rather than its output: an aggregate of services can read at full level while the base regenerating them is reduced beneath it, and only an observation reaching the base itself can detect the discrepancy.

Bias. Remark 3 shows that the loss need not be in the measurement: a biased indicator with an uncorrected certainty-equivalence policy empties the epistemic kernel of a system whose perfect-information kernel is nonempty. Monitoring therefore has two obligations — bounding the observation error and supplying the correction — and the second is the one governance structures can omit: the indicator is published, the correction is not applied, and the kernel empties although the information was sufficient.

Institutions. The hierarchy of Section 2.1 records that epistemic contraction is one of three causes of kernel shrinkage, the others being disturbances and institutional restriction (authority, enforcement, allocation). The epistemic-institutional kernel $\mathrm{IRViab}_{\mathcal{J}}(\mathcal{V})$ (Section 2.1) combines them: an institution restricted in what it may observe *and* in what it may command inherits both contractions. The obstruction calculus characterizes the first; its certificates remain valid, and typically tighten, when the institution's command set is further restricted, since restriction of the action correspondence can only empty kernels, never create them (one-sided monotonicity of viability under correspondence restriction).

Proposition 5 (monotonicity of the epistemic kernel)**.** The robust epistemic kernel is monotone in its constituents: enlarging the action sets, reducing the disturbance sets, refining the information structure, or enlarging the policy class never shrinks the kernel. Writing $\mathrm{ERViab}^{\cdot}$ for the kernel with the indicated constituent, $U_1 \subseteq U_2$, $D_1 \subseteq D_2$, $\Pi_1 \subseteq \Pi_2$, and "$\mathcal{I}_1$ refines $\mathcal{I}_2$" imply respectively

$$\mathrm{ERViab}^{U_1} \subseteq \mathrm{ERViab}^{U_2}, \qquad \mathrm{ERViab}^{D_2} \subseteq \mathrm{ERViab}^{D_1},$$

$$\mathrm{ERViab}^{\Pi_1} \subseteq \mathrm{ERViab}^{\Pi_2}, \qquad \mathrm{ERViab}^{\mathcal{I}_2} \subseteq \mathrm{ERViab}^{\mathcal{I}_1}.$$

Proof. Each inclusion is by restriction: a policy that is viable under the smaller action sets, the larger disturbance sets, the smaller policy class, or the coarser information structure remains admissible and nonanticipative under the corresponding enlargement or refinement, while the adversary's options are unchanged or reduced, so the same policy witnesses viability. $\square$

6.5 Limitations

The certificates are sound but not complete; several extensions are noted as open.

(i) The continuous-time certificates of Sections 3.1–3.4 are sufficient conditions for nonviability, not necessary-and-sufficient characterizations of the epistemic kernel (backward recursion is complete in the finite-horizon finite-system setting, Section 3.5); the middle ground (neither a Veliov-type condition nor an obstruction certificate) is open. In continuous settings completeness would additionally require a measurable-selection theory beyond the explicit selections the certificates exhibit, which is why completeness is claimed only where backward recursion is exact (Section 3.5).

(ii) The information-set semantics is set-membership; probabilistic or belief-space formulations require separate statements, although the common-action obstruction transfers mutatis mutandis to any semantics in which the policy must choose a single action per information state.

(iii) The results are finite-dimensional and time-invariant; the delay-free setting makes the timing bound of Theorem 3 sharp, and delay systems require the retarded extension of the Dini comparison argument (Hale and Verduyn Lunel, 1993).

(iv) Theorem 1's adversarial-realization step presupposes both the closed-graph regularity of Section 2.1 and the closed-loop existence hypothesis (H1.2) — D lower semicontinuous or constant, or the convexified reading; without these the certificate remains a heuristic.

(v) The institutional consequences of Section 6.4 are design conclusions from the certificates, not empirical findings; their empirical testing belongs to applied studies.

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7. Conclusion

Viability under perfect measurement has, under standard finite-dimensional regularity assumptions, a mature kernel, tangency, and approximation theory. Viability under incomplete observation has long been asymmetric: the sufficiency side is carried by Veliov's output-feedback condition and the estimation-tube programme, while the failure side lacked a comparable instrument. This paper supplies the missing instrument: the obstruction calculus develops sound, computationally interpretable certificates for nonviability — sufficient conditions for nonviability, equivalently necessary conditions for viability. They do not exhaust the complement of the epistemic kernel (Section 6.5), and the paper claims no complete characterization. The mechanisms are organized by one selector principle — observation-based viability is the existence of a common recursively safe response — and are nested in an obstruction ladder (Proposition 3); in finite systems backward belief recursion is sound and complete (Theorem 4), so the calculus is exact where it can be and sound elsewhere.

With displayed proofs and explicit witnesses, the certificates identify five mechanisms by which an observation structure can make robust viability impossible, plus one through which the policy class alone empties the kernel: the disturbance can enforce exit in finite time against every control; a constant observation can merge states with incompatible admissible controls (Proposition 1's admissibility form); the compatible states can demand incompatible actions at a single information state; the information can arrive after the enforced exit; and the observation fibres can cross the safe-set boundary, defeating every exact certifier — with certainty-equivalence control as the sixth, policy-class, mechanism. Each certificate that admits a design remedy identifies it: a faster indicator, a separating observation, an observation that separates the sides of each floor instead of aggregate certification, or the correction of a known bias. For sustainability governance, where observation is conducted through coarse indicators at discrete reviews, the calculus states the quantitative limits of what monitoring can deliver — and converts each impossibility into a design specification.

Appendix A: Bounded Constructions and Scope Remarks

The following bounded constructions complement the obstruction theorems of Sections 3–4. They are stated in full with their scope remarks.

A.1 Example (coupling creates viability absent in a factor). Take the two-patch logistic system $\dot{S}_i = g_i(S_i) + \kappa\,(S_j - S_i) - h_i, \quad i, j \in \{1, 2\},\ i \neq j$, with $g_i(s) = s(1 - s)$, $\kappa = 0.2$, constraints $S_i \in [0, 1]$ and $h_i \geq h_{\min,i}$ (harvest floors), and admissible controls the constant harvest vectors $h = (h_1, h_2)$ with $h_i \geq 0$. Choose $(S_1^*, S_2^*) = (0.5, 0.8)$. Define harvest floors by the equilibrium equations: $h_{\min,1} = g_1(0.5) + 0.2(0.8 - 0.5) = 0.31$; $h_{\min,2} = g_2(0.8) + 0.2(0.5 - 0.8) = 0.10$. Patch 1 in isolation: $\max_s g_1(s) = 0.25 < h_{\min,1} = 0.31$, so patch 1 is not viable in isolation. Yet the coupled system has the equilibrium $(0.5, 0.8)$ with $h = (0.31, 0.10)$ admissible and the stock constraints satisfied, so its kernel is nonempty. The example is the constructive counterpart of the emptiness results of Section 3: where the obstruction theorems certify that no policy works for reasons of information, patch coupling exhibits the opposite direction — a jointly viable operating point that no factor sustains alone.

A.2 Counterexample (emptiness despite factorwise viability at MSY). Take $\kappa > 0$, $C_1 \neq C_2$, $h_{\min,i} = r_i C_i/4$ (MSY level), in the same coupled system with $\phi_i(S_i) := g_i(S_i) - h_{\min,i}$ and growth laws $g_i(S) = r_i S(1 - S/C_i)$, so that

$$\phi_i(S_i) = -\frac{r_i}{C_i}\left(S_i - \frac{C_i}{2}\right)^2 \leq 0$$

with equality only at $S_i = C_i/2$, and the constraint set is the box $[C_1/2, \bar{S}_1] \times [C_2/2, \bar{S}_2]$. Each isolated system has kernel $[C_i/2, \bar{S}_i]$ (from $S_i > C_i/2$ the stock declines toward $C_i/2$

asymptotically, never below). For the coupled system, define the Lyapunov function $W = S_1 + S_2$; then

$$\dot{W} = \phi_1(S_1) + \phi_2(S_2) \leq 0,$$

with equality exactly at $p^* = (C_1/2, C_2/2)$, where the flow is nevertheless nonzero: $\big(\dot{S}_1, \dot{S}_2\big)\big|_{p^*} = \big(\frac{\kappa}{2}(C_2 - C_1), \frac{\kappa}{2}(C_1 - C_2)\big) \neq 0$. Suppose a trajectory remained in the constraint box for all time. Then W is non-increasing and bounded below by $(C_1 + C_2)/2$, so $W(t) \downarrow W_\infty \geq (C_1 + C_2)/2$; the ω-limit set of the trajectory is nonempty, compact, and invariant, and $\dot{W} \equiv 0$ on it (otherwise W would keep decreasing along the set). Hence the ω-limit set is contained in $\{p^*\}$, so it equals $\{p^*\}$ and the trajectory converges to p^*. But convergence to p^* is impossible: along a trajectory with $x(t) \to p^*$ the velocity satisfies $\dot{x}(t) = f(x(t)) \to f(p^*) \neq 0$, so $x(t) - p^* \sim (t - t_0)f(p^*)$, contradicting convergence. Therefore no trajectory remains admissible for all time: the kernel is empty, and since the box is compact and the vector field smooth, every trajectory violates the constraint in finite time. Note the logical order: the emptiness conclusion here rests on the Lyapunov–ω-limit argument, *not* on the absence of an equilibrium — absence of equilibria alone does not imply an empty kernel (periodic and nonstationary viable trajectories are possible in general), and the two are kept separate. The counterexample is specific to the MSY parameter choice; for $H_{\min,i} < r_i C_i/4$, equilibria may exist. Together with A.1 it delimits what coupling can and cannot repair: factorwise nonviability is repairable (A.1), but factorwise viability at an incompatible operating point is not (A.2).

References

Abaee, A.: The limits of compensatory aggregation: a formal separation of weak and strong sustainability assessment. Zenodo. https://doi.org/10.5281/zenodo.22545740 (2026).

Aubin, J.-P.: Viability Theory. Birkhäuser, Boston (1991)

Aubin, J.-P., Bayen, A.M., Saint-Pierre, P.: Viability Theory: New Directions, 2nd edn. Birkhäuser, Boston (2011)

Aubin, J.-P., Frankowska, H.: Set-Valued Analysis. Birkhäuser, Boston (1990)

Béné, C., Doyen, L., Gabay, D.: A viability analysis for a bio-economic model. Ecol. Econ. **36**, 385–396 (2001)

Cardaliaguet, P., Quincampoix, M., Saint-Pierre, P.: Differential games through viability theory: old and recent results. In: Jørgensen, S., Quincampoix, M., Vincent, T.L. (eds.) Advances in Dynamic Game Theory. Annals of the International Society of Dynamic Games, vol. 9, pp. 3–35. Birkhäuser, Boston (2007)

De Lara, M., Doyen, L.: Sustainable Management of Natural Resources: Mathematical Models and Methods. Springer, Berlin (2008)

Doyen, L., Gajardo, P.: Sustainability standards, multicriteria maximin, and viability. Nat. Resour. Model. **33**(3), e12250 (2020)

Doyen, L., Thébaud, O., Béné, C., Martinet, V., Gourguet, S., Bertignac, M., Fifas, S., Blanchard, F.: A stochastic viability approach to ecosystem-based fisheries management. Ecol. Econ. **75**, 32–42 (2012)

Farkas, J.: Theorie der einfachen Ungleichungen. J. Reine Angew. Math. **124**, 1–27 (1902)

Frankowska, H.: Optimal trajectories associated with a solution of contingent Hamilton–Jacobi equations. Appl. Math. Optim. **19**, 291–311 (1989)

Gale, D.: The Theory of Linear Economic Models. McGraw-Hill, New York (1960)

Haddad, G.: Monotone viable trajectories for functional-differential inclusions. J. Differ. Equ. **42**, 1–24 (1981)

Hale, J.K., Verduyn Lunel, S.M.: Introduction to Functional Differential Equations. Springer, New York (1993)

Maghenem, M., Sanfelice, R.G.: Characterizations of safety in hybrid inclusions via barrier functions. In: Proceedings of the 22nd ACM International Conference on Hybrid Systems:

Computation and Control (HSCC), pp. 109–118 (2019)
Martinez-Alier, J., Munda, G., O'Neill, J.: Weak comparability of values as a foundation for ecological economics. Ecol. Econ. **26**(3), 277–286 (1998)
Prajna, S., Jadbabaie, A.: Safety verification of hybrid systems using barrier certificates. In: Alur, R., Pappas, G.J. (eds.) Hybrid Systems: Computation and Control (HSCC 2004). LNCS, vol. 2993, pp. 477–492. Springer, Berlin (2004)
Prajna, S., Jadbabaie, A., Pappas, G.J.: A framework for worst-case and stochastic safety verification using barrier certificates. IEEE Trans. Autom. Control **52**(8), 1415–1428 (2007)
Prajna, S., Rantzer, A.: On the necessity of barrier certificates. IFAC Proc. Vol. **38**(1), 526–531 (2005)
Quincampoix, M., Veliov, V.M.: Viability with a target: theory and applications. In: Control Theory, Multivariate Analysis and Applications, pp. 47–63. Springer (1994)
Saint-Pierre, P.: Approximation of the viability kernel. Appl. Math. Optim. **29**, 187–209 (1994)
Veliov, V.M.: Sufficient conditions for viability under imperfect measurement. Set-Valued Anal. **1**, 305–317 (1993)
Witsenhausen, H.S.: A counterexample in stochastic optimum control. SIAM J. Control **6**(1), 131–147 (1968)

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